QR-008 PPE and handling by chemical

Doc ID	QR-008
Title	PPE and handling by chemical
Parent	CS-003 §6; CS-004 §6; CS-005 §6; CS-006 §5; CS-006 §6; CS-007 §6; CS-008 §6; CS-009 §6; CS-010 §6; CS-011 §5; CS-011 §6; CS-014 §6
Checked against	CS-003 Rev E; CS-004 Rev E; CS-005 Rev F; CS-006 Rev D; CS-006 Rev D; CS-007 Rev D; CS-008 Rev E; CS-009 Rev D; CS-010 Rev D; CS-011 Rev E; CS-011 Rev E; CS-014 Rev A

Derived from its parent standards and the safety data sheets they cite in `03 Datasheets/`. If this and a parent disagree, the parent is right.

1. What PPE do I need for any epoxy resin or hardener?

Nitrile gloves, safety glasses with side shields (EN 166), ventilation, and soap (never solvent) on skin, for every resin and hardener the team stocks (CS-008 §6). All of them can make you allergic through the skin (H317), and every hardener burns skin and can blind (H314, H318).

Also called: epoxy PPE, resin safety, safety gear, gloves, nitrile, safety glasses, goggles, SDS, MSDS, safety sheet, sensitization, allergy.

- Gloves: nitrile, always, on both components and on mixed resin. Change them as soon as resin gets on them (CS-004 §6, CS-006 §6).
- Eyes: safety glasses with side shields whenever resin or hardener is open. Prescription glasses are not eye protection (CS-004 §6).
- Ventilation: work in the ventilated shop area or with doors open, not in a closed side room (CS-004 §6).
- Respirator: for long or enclosed sessions with any resin, wear the team respirator with the stocked A + P100 cartridges (CS-008 §6; CS-004 §6, CS-007 §6); IN2 and AT30 should get ABEK instead (CS-006 §6). Cartridges by job: QR-008 §14.
- No resin or hardener SDS on file classifies it as a respiratory sensitizer (no H334); the XCR SDS §11.1 still warns the resin may be one (CS-008 §6). Skin sensitization adds up: repeated careless exposure can make you permanently allergic to epoxies, which ends your composites career (CS-004 §6).
- If you have ever had a skin reaction to epoxy, tell a lead. You shall not mix or brush XCR (CS-004 §6).

QR-008 §2 to §5 add what each system's SDS asks for on top of this.

2. IN2 infusion resin and AT30 hardener: gloves, glasses, respirator

Nitrile gloves at all times once resin is out, and safety glasses with side shields (EN 166) whenever IN2 or AT30 is open (CS-006 §6). In a closed room or a long session, wear the team respirator with the stocked A + P100 cartridges (CS-006 §6; CS-008 §6). You should use ABEK cartridges (EN 14387) instead; they are on the purchase list.

Also called: IN2, infusion resin, infusion epoxy, AT30, AT30 SLOW, AT30 FAST, slow hardener, fast hardener, Easy Composites resin, EC resin.

What the SDS says (`03 Datasheets/EC-SDS-IN2-Epoxy-Infusion-Resin-and-Hardener.pdf`, quoted in CS-006 §6):

- IN2 resin irritates skin and eyes, causes allergic skin reactions and may damage fertility (H315, H319, H317, H360F).

- AT30, SLOW and FAST, is corrosive: severe skin burns and serious eye damage (H314, H318), allergic skin reactions (H317), harmful if swallowed (H302).
- AT30 FAST adds possible organ damage through repeated exposure (H373) and is corrosive to the respiratory tract (EUH071).

Handling (CS-006 §6):

- Change gloves after any contamination.
- Mix and infuse in the ventilated shop area; the SDS asks for effective ventilation in all processing areas. Sensitization is cumulative.
- IN2 may damage fertility (H360F). Anyone who is pregnant, trying to conceive or concerned should get medical advice before handling it (SDS P201, P308 + P313), and may ask a lead for other work without saying why.
- The shop stocks A + P100 cartridges (CS-011 §5), not ABEK, so A + P100 is required and ABEK is a should.
- Cut spiral-wrap ends pierce gloves and bags alike. Tape them.
- A full mixed pot of IN2 left standing overheats (CS-006 §6; QR-008 §12).

3. XCR mold sealer: gloves, glasses and the respirator

Nitrile gloves always, safety glasses with side shields whenever XCR is open, and ventilation (CS-004 §6). For prolonged brushing on large molds, or poor ventilation, wear the team respirator with A1 (brown) cartridges. Seal in the open shop area or with doors open.

Also called: XCR, mold sealer, sealer, sealing resin, coating resin, epoxy coating resin, mould sealer.

What the SDS says (`03 Datasheets/EC-SDS-XCR-Epoxy-Coating-Resin-Combined.pdf`, quoted in CS-004 §6): resin and hardener both cause allergic skin reactions (H317) and serious eye damage (H318). The hardener causes severe skin burns (H314). The resin is suspected of causing genetic defects (H341). Neither is classified as a respiratory sensitizer (no H334), though SDS §11.1 says the resin may be one; the respirator is for vapor.

Rules (CS-004 §6):

- Anyone who has had a skin reaction to epoxy shall not mix or brush XCR, and tells a lead. There is other work on a sealing night.
- Gloves: nitrile (chemical-resistant per EN ISO 374-1), on both components and mixed resin. Change them if resin gets on them.
- Respirator: A-type (brown, EN 14387 A1) organic-vapor cartridges for undertray-scale molds or poor ventilation. Dust filters alone do nothing against epoxy vapors.
- Eyes: you should wear safety goggles with side protection (EN 166) when mixing or brushing XCR. Goggles are on the purchase list; until then, safety glasses with side shields.
- Skin: wash any contact at once with soap and water, not solvent. Contaminated clothing stays at the shop (XCR SDS P272).

4. West System 105 and 205, 206, 209 hardeners: gloves, glasses, respirator

Nitrile gloves, safety glasses with side shields, ventilation and no solvent on skin; a respirator with an A-filter for large overhead or enclosed laminating (CS-007 §6, CS-008 §6). The hardeners are worse than the resin: all three burn skin and damage eyes.

Also called: West, West System, 105, 205, 206, 209, fast hardener, slow hardener, extra slow hardener, wet layup epoxy, laminating resin.

What the SDSs say (`03 Datasheets/WEST-SDS-105-Epoxy-Resin.pdf`, `WEST-SDS-205-Fast-Hardener.pdf`, `WEST-SDS-206-Slow-Hardener.pdf`, `WEST-SDS-209-Extra-Slow-Hardener.pdf`, quoted in CS-007 §6 and CS-008 §6):

Product	Hazards for people
105 resin	H315 skin irritation, H317 allergic skin reaction, H319 eye irritation
205 fast hardener	H302, H312, H314, H317, H318, H341 suspected of causing genetic defects, H373 organ damage through repeated exposure
206 slow hardener	H302, H314, H317, H318, H335 may irritate the lungs
209 extra slow hardener	H302, H314, H317, H318, H335, H373

Handling:

- Wash skin with soap, never solvent (CS-008 §6).
- 105 with 205 in a pot gets hot fast. The exotherm rules apply (CS-007 §6; QR-008 §12).
- Treat 105 and its hardeners the way CS-008 §6 treats every resin system (QR-008 §1).

5. DP420 structural adhesive: gloves, glasses, respirator

Nitrile gloves and safety glasses with side shields are required for DP420 mixing and bonding, and a respirator with OV/P100 cartridges when ventilation is not adequate (CS-009 §6). You should also wear indirect vented goggles and a full face shield, both on the purchase list.

Also called: DP420, DP 420, 3M DP420, Scotch-Weld, structural adhesive, structural epoxy, epoxy glue, two-part glue, cartridge glue, duo-pak.

Why: Part A causes severe skin burns and eye damage, which is why its SDS recommends goggles and a face shield (CS-009 §6, `03 Datasheets/3M-SDS-DP420-Black.pdf` Part A §8.2.2). The SDS rates nitrile and butyl gloves over 4 h for both parts (§8.2.2).

If Part A gets on you (DP420 SDS, Part A §4):

1. Skin: flush at once with large amounts of water for at least 15 minutes.
2. Take off contaminated clothing.
3. Get immediate medical attention.
4. Eyes: flush at once with large amounts of water for at least 15 minutes, removing contact lenses if that is easy, and get medical attention immediately.

Bond prep and the DP420 steps are in QR-013.

6. Acetone and IPA: which gloves, and how long they last

Nitrile gloves and safety glasses with side shields are required for a solvent wipe (CS-009 §6). Thin nitrile (0.11 mm) holds 100% IPA for 117 min, but acetone gets through even Ansell's 0.45 mm latex/neoprene/nitrile glove in 17 min (ASTM F739; 21 min by EN ISO 374): change gloves as soon as acetone wets them.

Also called: acetone, IPA, isopropyl alcohol, isopropanol, rubbing alcohol, 91% IPA, solvent, solvent wipe, cleaner, glove breakthrough, permeation.

Where the numbers come from: Ansell's charts, `03 Datasheets/ANSELL-chart-TouchNTuff-92-500-93-250.pdf` (0.11 mm nitrile) and `03 Datasheets/ANSELL-chart-TouchNTuff-93-800.pdf` (0.45 mm NRL/neoprene/nitrile), quoted in CS-009 §6. They test Ansell's gloves; the team's brand may differ.

- Respirator: none set. The solvent SDSs on file are stand-ins, so no number here comes from them (CS-009 §6).
- Wipe with ventilation, and put out ignition sources while solvents are open (CS-009 §6, §7.3).
- Acetone and 91% IPA are flammable (CS-014 §6). Acetone lives in the flammables cabinet (CS-011 §6). CS-011 §6 does not name IPA, and CS-004 §5 lists the 91% bottle in the Chemicals box.

- Never acetone on or near tooling board (CS-009 §7.3).
- On molds, use the 91% isopropyl bottle, never the 70% rubbing alcohol (CS-004 §5).
- Never use solvent to get resin off skin. Soap and water (CS-008 §6).

7. Sanding and cutting dust: carbon, glass, G10 and tooling board

Cutting, sanding or machining cured CF, glass or G10 requires the dust extractor running at the cut, a half-facepiece respirator with OV/P100 cartridges, nitrile gloves, long sleeves, safety glasses with side shields, and a second trained person present (CS-009 §6).

Also called: dust, carbon dust, CF dust, fibre dust, fiber dust, sanding, cutting, trimming, Dremel, dust mask, P100, respirator, extractor, tooling board dust, board dust.

- CF dust is the team's worst routine exposure (CS-009 §6).
- Cut CF edges splinter. Gloves are for that as much as for the dust.
- Supervision (set in SN5; the Composites Lead may change it): no cutting cured carbon without a second trained person present. Trained means holding the Trimming and finishing training in the app (CS-009 §6). The lead adds that training in People → Catalog; until a lead has added and granted it, ask the lead who counts as trained.
- The shop stocks A + P100 cartridges; 3M 60921 is the OV/P100 cartridge on file (CS-009 §6, `03 Datasheets/3M-spec-60921-OV-P100-Cartridge.pdf`).
- Tooling-board dust (gluing prep, hand-shaping) is abrasive and irritating: a dust mask at minimum, a respirator and extraction for power sanding (CS-003 §6).
- On the ShopSabre: extraction on, a dust mask at minimum, a respirator for long jobs (CS-005 §6).
- Vacuum shared rooms. Do not blow dust around with compressed air (CS-005 §6).

Trimming and sanding steps are in QR-013.

8. Ure-Bond 90 tooling board glue: ventilation, gloves, respirator (isocyanate)

Glue only with room-size ventilation, and never heat or spray Ure-Bond 90. Wear safety glasses with side shields, long sleeves and gloves (CS-003 §6). Part A contains MDI, an isocyanate that may cause allergy or asthma symptoms if inhaled (H334). If the room cannot be ventilated, wear the CS-009 respirator (CS-003 §6).

- Asthma or a past reaction to isocyanates: you shall not mix Ure-Bond 90. Tell a lead; there is other work on a gluing night (CS-003 §6).
- Gloves: the SDS names liquid-tight butyl rubber, neoprene or PVC gloves, and the technical bulletin says rubber gloves; neither names nitrile. Until the shop stocks one of those, wear the shop's nitrile gloves. You should wear butyl, neoprene or PVC gloves; they are on the purchase list (CS-003 §6).

Also called: Ure-Bond, Urebond, Ure Bond 90, board glue, tooling board adhesive, urethane adhesive, Smooth-On, MDI, isocyanate, gluing the stack.

What the SDS says (`03 Datasheets/1386A_2-1386B_2.pdf`, quoted in CS-003 §6): Part A contains MDI (methylene diphenyl diisocyanate, CAS 101-68-8, 15 to 35 %). It causes skin and eye irritation, may cause an allergic skin reaction, is harmful if inhaled, may cause allergy or asthma symptoms if inhaled, may irritate the airways, is suspected of causing cancer and may damage the airways through repeated exposure (H315, H319, H317, H332, H334, H335, H351, H373). Part B is not classified.

- The technical bulletin warns the vapors become significant when heated or sprayed (`03 Datasheets/URE_BOND_90_TB.pd` CS-003 §6).
- Eyes: flush with water for 15 min and get medical attention (TB).
- Skin: remove with soap and water (TB).
- Clamped assemblies stay put for 4 h (TB).

9. Clear coat (LiME LiNE UC4:1): Part B stays shut

Nobody opens, mixes or applies Part B, the isocyanate activator, until its SDS or the can's hazard text is in `03 Datasheets/` and CS-009 §6 quotes its PPE. For Part A, a respirator with OV/P100 cartridges and nitrile gloves are required (CS-009 §6).

Also called: clear coat, clearcoat, clear, LiME LiNE, Lime Line, UC4:1, urethane clear, activator, Part B, GH60, GH70, GH80.

Part A (`03 Datasheets/LIMELINE-SDS-UC4-1-Clearcoat-Part-A.pdf`, quoted in CS-009 §6):

- Hazards: highly flammable liquid and vapor (H225), may cause an allergic skin reaction (H317), serious eye irritation (H319), may cause drowsiness or dizziness (H336).
- Use it only outdoors or in a well-ventilated area.
- Eyes: you should wear tightly sealed goggles, which are on the purchase list. Until then, safety glasses with side shields.
- Respirator: the OV/P100 respirator covers brief exposure. For intensive or longer exposure the SDS asks for a respirator independent of circulating air; the team sets that when Part B's PPE is quoted (CS-009 §7.4 step 1).
- Gloves: nitrile. The SDS names no glove material.
- Part A is flammable (CS-014 §6) and lives in the flammables cabinet (CS-011 §6).

10. Airtac 2 spray adhesive: flammable, how to spray it safely

Airtac 2 is an extremely flammable aerosol: no sparks or heat, use it in a ventilated space, and spray in short bursts (CS-006 §6). The SDS on file, the 2005 MSDS of the earlier Airtac 2 Spray Adhesive (a current one is wanted, CS-006 R5), classes it as a Class IA flammable liquid with a flash point of −42 °F (CS-014 §6), so it is stored in the flammables cabinet (CS-011 §6).

Also called: Airtac, Air Tac, Airtac 2, tack spray, spray tack, spray adhesive, spray glue, Airtech spray.

- SDS: `03 Datasheets/airtac2spray.pdf`.
- Use a light mist only, from at least 30 cm (CS-006 §5).
- When it arrives, set the lot's Hazard to flammable (CS-014 §6).

11. Mold release wax, PVA, graphite powder and copper mesh

Release wax and PVA are low hazard: gloves and ventilation suffice (CS-004 §6). Graphite powder needs the respirator with a P100 cartridge and nitrile gloves while you weigh and mix it (CS-010 §6). Cut copper mesh has sharp wire ends: wear gloves to cut and place it.

Also called: wax, Formula Five, release wax, PVA, Partall, release film, graphite, graphite powder, conductive powder, copper mesh, grounding mesh.

- Wax and PVA: Formula Five Mold Release Wax and Partall Film #10 (CS-004 §5). Their SDSs are `03 Datasheets/SDS-F5MRW-5.3-en-NA-2025-01-28.pdf` and `03 Datasheets/SDS-PF10-5.5-en-NA-2025-01-28.pdf`.
- Graphite: the shop stocks A + P100 cartridges and nitrile gloves, so both are required (CS-010 §6). No graphite SDS is on file yet.
- Copper mesh: gloves when cutting and placing it (CS-010 §6).

12. A mixing cup is getting hot or smoking (exotherm)

A warm cup: use it at once or spread it thin. A hot or smoking cup: carry it outside onto concrete and leave it. Do not bin it and do not dilute it with water indoors. Nothing goes in the trash until it is fully cured and cool (CS-008 §6).

Also called: hot cup, hot pot, exotherm, smoking cup, runaway, cooking off, kicking off, cup going off.

Mass concentrated in a cup speeds the cure up, and the cure's own heat speeds it up again (CS-008 §4, §6). To stop it happening:

1. Mix the smallest batch that does the job (CS-008 §6).
2. Pour out of the pot promptly (CS-008 §6).
3. Spread leftovers thin on scrap, or leave them outside to cure (CS-008 §7 step 7).

Which systems bite fastest: 105 with 205 in a pot gets hot fast (CS-007 §6); mixed XCR left in the pot self-accelerates (CS-004 §6); a full mixed pot of IN2 left standing overheats (CS-006 §6).

13. Resin or hardener on skin or in an eye: first aid

In an eye: rinse with plenty of water, under the eyelids too, for at least 15 minutes, then get medical attention (CS-008 §6). On skin: wash it off at once with soap and plenty of water, never solvents or thinners (IN2 SDS §4).

Also called: first aid, splash, eye wash, eyewash, rinse, chemical burn, skin contact, rash, allergic reaction, dermatitis.

Eye:

1. Rinse with plenty of water at once.
2. Hold the eyelids open and rinse under them.
3. Take out contact lenses if that is easy, and keep rinsing (West 105 SDS, P305+351+338).
4. Keep rinsing for at least 15 minutes (CS-008 §6).
5. Get medical attention.

Skin:

1. Take off contaminated clothing (XCR SDS, P303+P361+P353). It stays at the shop (CS-004 §6).
2. Wash with soap and plenty of water. No solvent (`03 Datasheets/EC-SDS-IN2-Epoxy-Infusion-Resin-and-Hardener.` §4).
3. DP420 Part A on skin: flush for at least 15 minutes and get immediate medical attention (DP420 SDS, Part A §4; QR-008 §5).
4. A rash or other skin reaction afterwards: tell a lead. You shall not mix or brush XCR again (CS-004 §6).

14. Which respirator cartridge for which job?

Dust, DP420 in poor ventilation and clear coat Part A take OV/P100 cartridges; the shop stocks A + P100 (CS-009 §6, required). XCR and West vapor take A-type cartridges. AT30 should get ABEK, which is on the purchase list (CS-006 §6).

Also called: respirator, mask, cartridge, filter, organic vapor, OV, OV/P100, P100, A1, ABEK, 60921, half mask, half-facepiece.

Job	Cartridge	Rule	Source
Cutting, sanding or machining cured CF, glass, G10	OV/P100, half-facepiece	Required	CS-009 §6
DP420, ventilation not adequate	OV/P100	Required	CS-009 §6
Clear coat Part A	OV/P100	Required	CS-009 §6
Graphite powder	P100	Required	CS-010 §6
XCR, prolonged brushing on large molds or poor ventilation	A1 (brown)	Wear it	CS-004 §6

Job	Cartridge	Rule	Source
West, large overhead or enclosed laminating	A-filter	Wear it	CS-007 §6
IN2 and AT30, closed room or long session	A + P100	Required	CS-006 §6, CS-008 §6
IN2 and AT30, closed room or long session	ABEK	Should (purchase list)	CS-006 §6
Ure-Bond 90, room cannot be ventilated	The CS-009 respirator	Required	CS-003 §6
Solvent wipe (IPA, acetone)	None set		CS-009 §6

Required means PPE the shop stocks, which the standards make a shall. Should means PPE an SDS asks for that the shop does not stock yet; each of those items is on the purchase list (Simon, 2026-10-10).

Dust filters alone do nothing against epoxy vapors (CS-004 §6). The shop's restock list carries A + P100 cartridges (CS-011 §5).

15. Where do chemicals get stored, and what if one leaks?

Resins and hardeners go in the chemical boxes or barrel, never in a food fridge; flammables (Airtac, acetone, clear coat) go in the flammables cabinet (CS-011 §6). A leaking container never goes on a shelf: keep it in its packaging, set it apart and tell a lead (CS-014 §6).

Also called: chemical storage, storing resin, flammables cabinet, fridge, leak, spill, damaged container, unpacking a delivery.

Co-storage: see CS-011 §6; under review.

CS-011 §6 as it stands also says to segregate hardeners from resins on different shelves.

Unpacking a delivery (CS-014 §6):

1. Wear nitrile gloves to unpack and shelve chemical containers.
2. Before you touch a damaged or leaking container, read its SDS.
3. Leave a leaking container in its packaging, apart from other chemicals, and tell a lead. The lead decides what happens to it.
4. Put chemicals straight onto a location CS-011 §6 allows. Never a fridge.

Which site is which, and how to record a move, is in QR-014.